Practice workbook · Explain how the eye focuses on near and distant objects

Ciliary muscle, suspensory ligament and lens changes for near and distant focus, as tables, drawings, data-based or written explanations.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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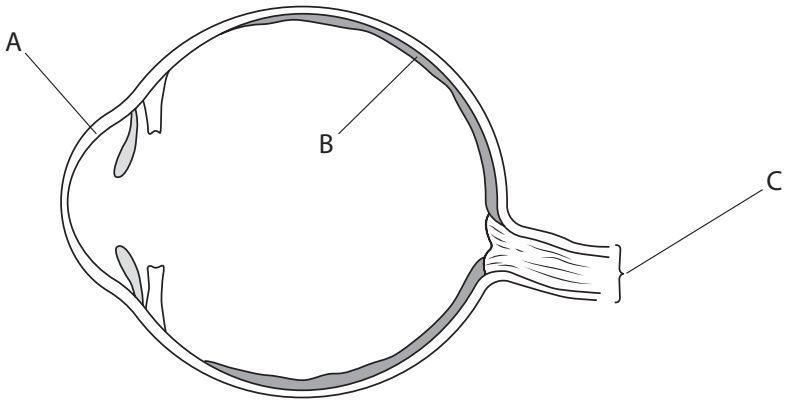
Type ID: HBI-EXPLAIN-LENS-ACCOMMODATION

1. Question 2 · part b · 4 marks

June 2025 Paper 1R

Worked-example retry · Attempt part b only.

2 The diagram shows a section through the eye with the lens missing.



(a) (i) Complete the diagram by drawing in the lens and suspensory ligaments as they would appear when the eye is looking at a near object.

(2)

(ii) Name the parts labelled A, B and C.

(3)

A .....

B .....

C .....

(b) A person is looking at a near object.

Describe the changes that occur in the eye if the person then looks at a distant object.

(4)

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(Total for Question 2 = 9 marks)

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1. Question 2 · part b · 4 marks · Working and answer

Working and answer

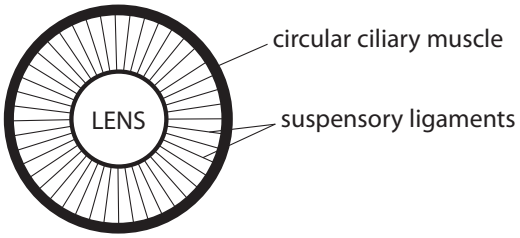
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2. Question 5 · part a · 3 marks

January 2022 Paper 2

Further practice · Attempt part a only.

5 (a) The diagram shows a front view of structures in the eye that help to form an image on the retina.



The three structures shown in the diagram change when focusing on near and distant objects.

Complete the table by putting ticks (✓) in the correct boxes to show the condition of each structure of the eye when focusing on near and distant objects.

(3)

	Focusing on a near object	Focusing on a distant object
Lens	thin ☐ thick ☐	thin ☐ thick ☐
Circular ciliary muscle	contracted ☐ relaxed ☐	contracted ☐ relaxed ☐
Suspensory ligament	stretched ☐ loose ☐	stretched ☐ loose ☐

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2. Question 5 · part a · 3 marks

January 2022 Paper 2

Further practice · Attempt part a only.

(b) Humans have stereoscopic vision.

Explain what is meant by the term **stereoscopic vision**.

(3)

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2. Question 5 · part a · 3 marks · Working and answer

Working and answer

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3. Question 4 · part b.ii · 1 mark

June 2023 Paper 1

Further practice · Attempt part b.ii only.

4 The eye contains receptors that can detect light.

(a) Name two other senses.

(2)

1 .....

2 .....

(b) The table shows how the focal length of the eye lens changes as the thickness of the lens changes.

Thickness of lens in mm	Focal length in mm
10	12.5
9	14.0
8	15.5
7	18.0
6	21.0
5	25.0

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(i) Describe the relationship between the thickness of the lens and its focal length.

(2)

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(ii) Give the effect on the diameter of the lens as its thickness increases.

(1)

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3. Question 4 · part b.ii · 1 mark

June 2023 Paper 1

Further practice · Attempt part b.ii only.

(iii) A person is looking at a distant object and then looks at a near object.

Use the data in the table to explain the changes that occur in the eye so the person is able to form an image of the near object on the retina.

(4)

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3. Question 4 · part b.ii · 1 mark · Working and answer

Working and answer

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4. Question 4 · part b.iii · 4 marks

June 2023 Paper 1

Further practice · Attempt part b.iii only.

4 The eye contains receptors that can detect light.

(a) Name two other senses.

(2)

1 .....

2 .....

(b) The table shows how the focal length of the eye lens changes as the thickness of the lens changes.

Thickness of lens in mm	Focal length in mm
10	12.5
9	14.0
8	15.5
7	18.0
6	21.0
5	25.0

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(i) Describe the relationship between the thickness of the lens and its focal length.

(2)

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(ii) Give the effect on the diameter of the lens as its thickness increases.

(1)

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4. Question 4 · part b.iii · 4 marks

June 2023 Paper 1

Further practice · Attempt part b.iii only.

(iii) A person is looking at a distant object and then looks at a near object.

Use the data in the table to explain the changes that occur in the eye so the person is able to form an image of the near object on the retina.

(4)

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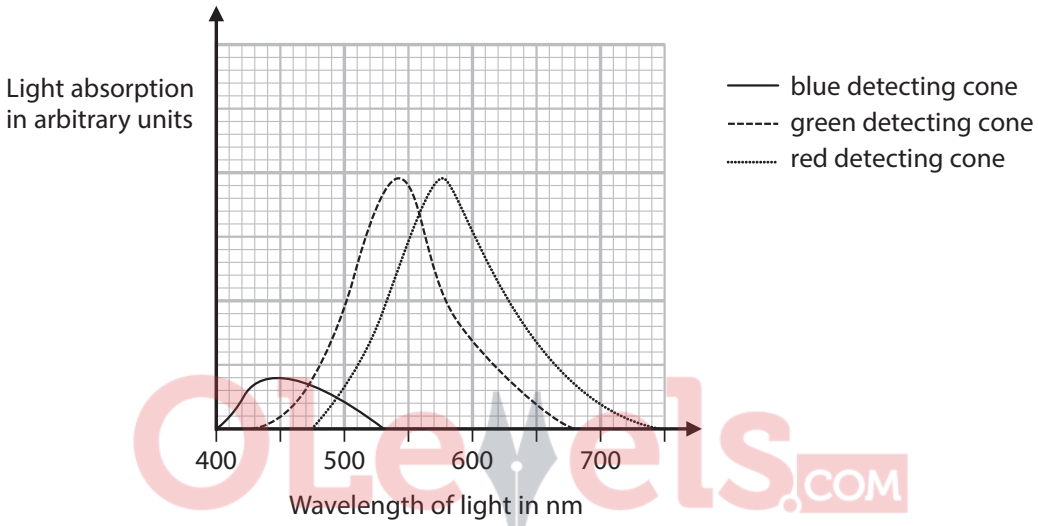
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4. Question 4 · part b.iii · 4 marks

June 2023 Paper 1

Further practice · Attempt part b.iii only.

(c) Cone cells are found in the retina of the eye. They detect colour. The graph shows the absorption of different wavelengths of light by three types of cone cell.



(i) Determine the wavelength of light at which a red detecting cone absorbs most light.

(1)

wavelength of light = ... nm

(ii) Determine the maximum difference in the wavelengths of the light that can be absorbed by blue detecting cones.

(2)

difference in wavelengths = ... nm

(iii) Suggest why it is better to have three types of cone rather than just one type of cone.

(2)

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(Total for Question 4 = 14 marks)

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4. Question 4 · part b.iii · 4 marks · Working and answer

Working and answer

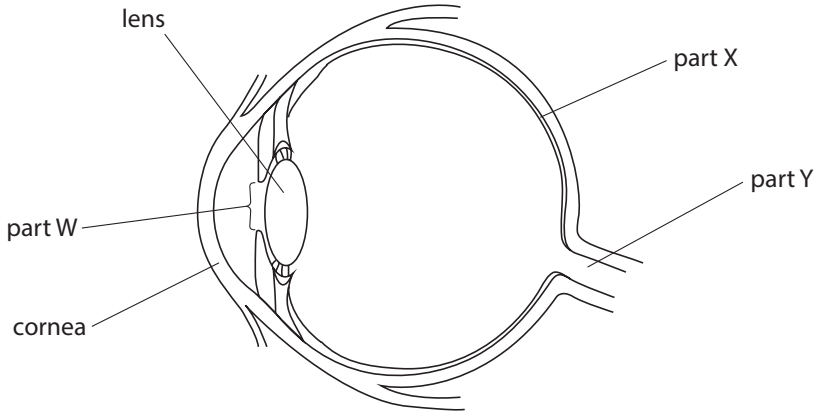
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5. Question 3 · part a.iii · 2 marks

November 2024 Paper 1

Further practice · Attempt part a.iii only.

3 (a) The diagram shows the human eye.



(i) Name part W and part X.

(2)

W .....

X .....

(ii) Describe the function of part Y.

(2)

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(iii) Explain the change in the shape of the lens when the eye views a distant object.

(2)

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5. Question 3 · part a.iii · 2 marks

November 2024 Paper 1

Further practice · Attempt part a.iii only.

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(iv) Describe how corneal transplants help to restore normal vision for people with corneal disease.

(2)

(b) The iris reflex controls the amount of light entering the eye.

Describe a method that could be used to investigate how the eye responds to different light intensities.

(4)

(Total for Question 3 = 12 marks)

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5. Question 3 · part a.iii · 2 marks · Working and answer

Working and answer

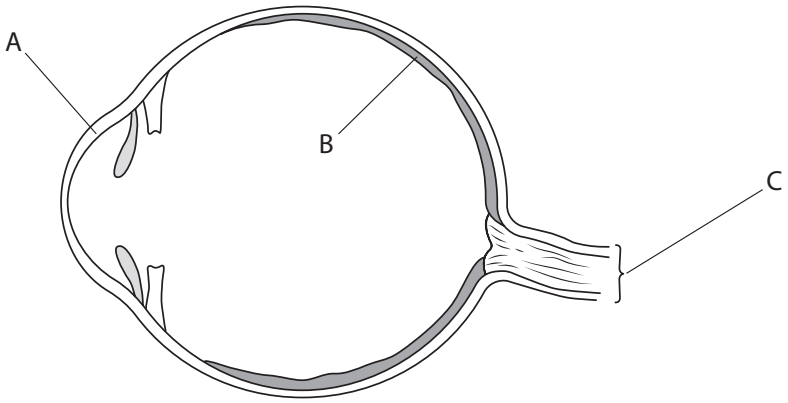
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6. Question 2 · part a.i · 2 marks

June 2025 Paper 1R

Further practice · Attempt part a.i only.

2 The diagram shows a section through the eye with the lens missing.



(a) (i) Complete the diagram by drawing in the lens and suspensory ligaments as they would appear when the eye is looking at a near object.

(2)

(ii) Name the parts labelled A, B and C.

(3)

A .....

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C .....

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(Total for Question 2 = 9 marks)

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6. Question 2 · part a.i · 2 marks · Working and answer

Working and answer

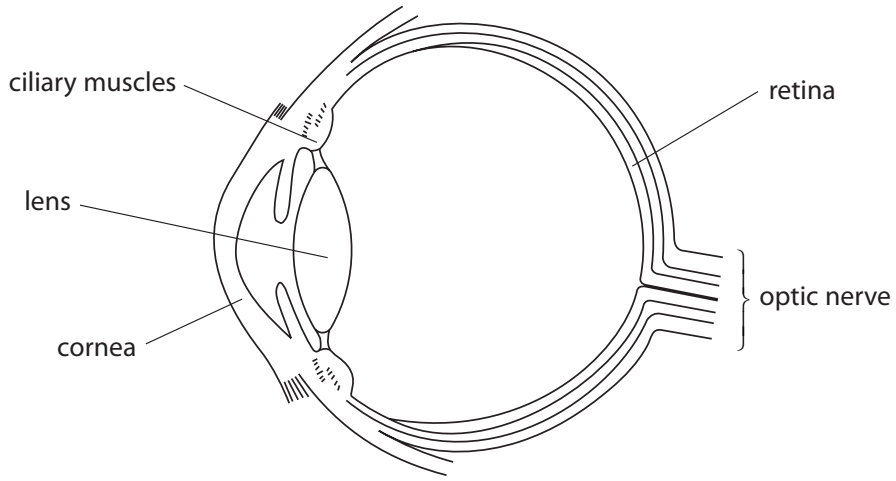
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7. Question 6 · part whole · 7 marks

June 2022 Paper 1R

Further practice · Attempt part whole only.

6 The diagram shows a section through the human eye.



Describe the role of the labelled structures in seeing a clear image of an object.

(Total for Question 6 = 7 marks)

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7. Question 6 · part whole · 7 marks · Working and answer

Working and answer

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